

Perfect Mode Hal: Aggressive, Change-Aware Opponent Exploitation in Exact DTH

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Abstract

Perfect Mode (PM) Hal is a causal meta-policy for repeated, pure Drop the Handkerchief (DTH). One certified tablebase remains the sole authority for each continuation-adjusted 60×60 matrix. Around it, PM combines Adaptive Hal’s Bayesian counts, Perfect Hal’s pattern experts, online change-point and outcome models, and a compatible Aggro Hal recurrent network. Fixed Share learns role-specific source and controller weights. Four modes move from an exact shield to hard forecast responses, while every selected policy receives a fresh local worst-case-loss measurement. An earlier 24-identity development run scored 44 wins in 48 games, but its protocol lacked durable registration and its evaluator trusted self-reported risk. It has no promotion authority. This note instead registers a 56-identity confirmation protocol with independent matrix recomputation, a no-Aggro ablation, and family and seat gates. At the protocol commit that confirmation has not been executed. PM is an aggressive candidate, not human mind reading, dominance, or a whole-game safety theorem.

1 Scope and design philosophy

At public state x , the exact DTH agent supplies matrix $M(x)$, equilibrium policies (x^*, y^*) , and a saddle-gap certificate at most 10^{-6} . All actions are literal seconds $1, \dots, 60$. PM never estimates continuation values and never sees an unrevealed simultaneous action.

The intended character is “overbearing after evidence, composed after surprise.” Perfect contributes hard exploitation; Adaptive contributes uncertainty and a constrained response frontier; Aggro contributes recurrent features and a learned residual policy. The story evidence ledger motivates competing personae, opponent mental models, and abrupt memory discontinuities. PM translates that fiction into testable engineering—model competition, role separation, recency, and change recovery—rather than treating a person as one fixed psychological type. The final narrative ideal, to improve and adapt without looking back, becomes persistent memory with rapid de-weighting and recovery after changed play.

2 Causal opponent model

For each opponent role separately, PM maintains six mandatory forecast sources: uniform recovery, the opponent’s exact equilibrium policy, a decayed Dirichlet model, Perfect’s online expert mixture, categorical Bayesian online change-point detection (BOCPD), and an outcome-conditioned model. A seventh source is Aggro’s GRU forecast when a strict compatible checkpoint is present. After action a_t is revealed, source k receives prequential log score and Fixed Share updates

$$\begin{aligned}\tilde{w}_{k,t+1} &\propto w_{k,t} \exp(\eta \log q_{k,t}(a_t)), \\ w_{k,t+1} &= (1 - \alpha)\tilde{w}_{k,t+1} + \frac{\alpha}{K - 1} (1 - \tilde{w}_{k,t+1}),\end{aligned}$$

with $\alpha = .04$. Thus every source retains recovery mass when an opponent switches. BOCPD carries a truncated Dirichlet–multinomial run-length posterior with hazard .08. The fused forecast is $\hat{q}_t = \sum_k w_{k,t} q_{k,t}$.

Aggression confidence is the declared heuristic

$$c_t = c_{\max} \frac{n_{\text{eff}}}{4 + n_{\text{eff}}} (1 - P_t(\text{change}))(1 - \text{JS}_t)s_t,$$

where JS_t is normalized source disagreement and $s_t = \exp[-\max(0, \overline{\text{NLL}}_t - \log 60)]$. Only the evidence fraction has a data-biased-response interpretation; the product is a project controller, not a published theorem.

3 Aggression with measured risk

For Dropper, the certified ϵ candidate solves

$$\max_{x \in \Delta_{60}} x^\top M \hat{q}_t \quad \text{s.t.} \quad (M^\top x)_j \geq L - \epsilon \quad \forall j,$$

where $L = \min_j (M^\top x^*)_j$ is freshly recomputed. Checker uses the symmetric upper constraint. The bank also contains exact equilibrium, hard responses to fused and Perfect forecasts, and Aggro’s direct policy. For a direct Dropper policy x , actual local loss is $\max(0, L - \min_j (M^\top x)_j)$; every direct and frontier candidate is independently checked against the current matrix.

One common bank of heuristic ensemble perturbations must give positive expected improvement and at least .55 positive-draw support. These draws are not a calibrated posterior, so the support fraction is not an improvement probability. Cold start or a change shock enters *shield*,

where only the exact face is admissible. The registered v3 caps are *probe* .05, *press* .50, and *dominate* 2.0, inside a 12.0 per-game charge budget. Controller weights receive full-information counterfactual payoff feedback after reveal. This is deliberately aggressive: the budget is an explicit risk appetite, not a guarantee that losses add across a DTH game. Ganzfried–Sandholm’s cumulative ledger concerns repetitions of one stage game; DTH’s matrices change with the state and already contain overlapping continuation values.

4 Development audit and registered confirmation

The v1 and v2 files were created and run before this feature branch registered them in Git. File status text and mutable modification times cannot establish preregistration. Both runs are therefore development evidence. The v2 run used 24 parameter identities across six opponent families. Each identity produced a common seeded seat pair for PM, Perfect, Adaptive, Aggro, and Exact; family plus parameter seed was the bootstrap unit (10,000 replicates). Stopped games scored .5.

Table 1: V2 development outcomes (48 games per controller).

Controller	Score	W–L–S	Mean halves
PM Hal	.9167	44–4–0	6.63
Perfect	.8958	43–5–0	5.56
Adaptive	.6563	31–16–1	7.10
Aggro	.6458	31–17–0	7.23
Exact	.5417	26–22–0	9.46

Paired PM-minus-baseline score differences were +.3750 [.2083, .5417] versus Exact, +.2604 [.1354, .3854] versus Adaptive, +.2708 [.1042, .4375] versus Aggro, and +.0208 [−.0417, .0833] versus Perfect. Those intervals are descriptive. The run cannot promote PM, and even its nominal gate failed the nonnegative lower-bound condition against every component.

Across 318 PM decisions, shield/probe/press/dominate counts were 56/46/154/62. The controller reported maximum local loss .2561 and no understated-loss or game-budget violations, but the v2 evaluator did not independently reconstruct every selected policy’s loss. PM’s identity-clustered expected NLL was 3.481 [2.904, 3.978]. Fused-minus-source NLL was −.252 [−.341, −.162] against Perfect and −.614 [−1.184, −.119] against uniform; negative is better. The fused forecast beat every individual source on this metric. Change detection remained weak: it detected 10 of 86 synthetic truth shifts, with one false-alarm onset. Strong play did not validate that detector.

The v3 confirmation is registered in Git before execution. It binds the controller, recurrent checkpoint, and tablebase by SHA-256; requires a clean worktree; records the executing commit; and assigns four new seeds to all 14 registered synthetic families, for 56 independent identities and 112 games

per controller. It compares PM with Exact, Perfect, Adaptive, Aggro, and PM without Aggro. Promotion requires zero independent risk-audit violations, nonnegative clustered lower bounds against Exact, every component, and the no-Aggro ablation, plus nonnegative point differences in both seats and every family. The evaluator reloads every continuation-adjusted matrix and recomputes selected worst-case loss, charge contracts, mode caps, cumulative charges, and per-game budgets. No v3 result appears in this protocol-registration version of the note.

5 Human-play interpretation and limits

The psychology is operational: reinforcement, belief learning, response to one’s own revealed action, recency, periodicity, and change are competing causal hypotheses. A human can counter-adapt, deceive, or change strategy; PM must preserve recovery mass and retreat on surprise. It should never label a participant’s personality or claim privileged access to intent.

A human study must keep opponent identity across games, log the sequence forecast before each reveal, score realized NLL/Brier without simulator-truth distributions, and cluster outcomes by participant. Change delay and false alarms, role slices, calibration, and game score are co-primary diagnostics. Bounded recursive reasoning is one possible population archetype, not a label the controller may assign permanently to a participant. The present measured evidence is synthetic development evidence from 24 identities and known families. The recurrent checkpoint saw 2,653 training decisions; its causal contribution remains an open question until the registered no-Aggro ablation is read. PM is feature-complete for human play as an aggressive candidate, while promotion awaits the registered synthetic result and then participant-level human evidence.

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